DATA SCIENCE & ARTIFICIAL INTELLIGENCE

& CS/IT

Calculus and Optimization

Functions and Graphs

Lecture No. 03



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Recap of previous lecture



Topic

FUNCTIONS & THEIR GRAPHS - 2

Topics to be Covered



Topic

FUNCTIONS & THEIR GRAPHS - 3

RECAP:

Function: if $\forall x \in A$ $\exists$ unique $y \in B$ s.t $y = f(x)$ then f is called func^n

(y: output, x: Input)

Domain: set of permissible Inputs for f is called Domain.

Range " " outputs " " " Range. func^n.

V-line Test → Line $\perp^r$ to y axis cuts the graph only at one point then it is a

Even func^n: if $f(-x) = f(x)$ Symmetry about y axis occurs

Odd func^n: if $f(-x) = -f(x)$ " " origin occurs

NEITHER func^n: if $f(-x) \neq f(x)$ & $\neq -f(x)$

Periodic func^n: if $f(x+T) = f(x)$ i.e it's graph Repeats after Period 'T'

RECAP

Types of functions

- ALGEBRAIC function
  - (1) Polynomial func^n
  - (2) Rational func^n
  - (3) Irrational func^n
  - (4) Piecewise func^n
    - Mod func^n
    - Sgn func^n.
    - G.I.f.
    - L.I.f
    - F.P.f
- TRANSCEDENTAL function
  - (1) Exponential func^n
  - (2) log function.
  - (3) Trigonometric func^n
  - (4) Inverse Trig. functions.

Sgn = SIGNUM. Func^n.

G.I.f = Greatest Integer Func^n (Floor func^n)

L.I.f = Least Integer func^n (Ceiling func^n)

F.P.F = Fractional Part Func^n.

RECAP:

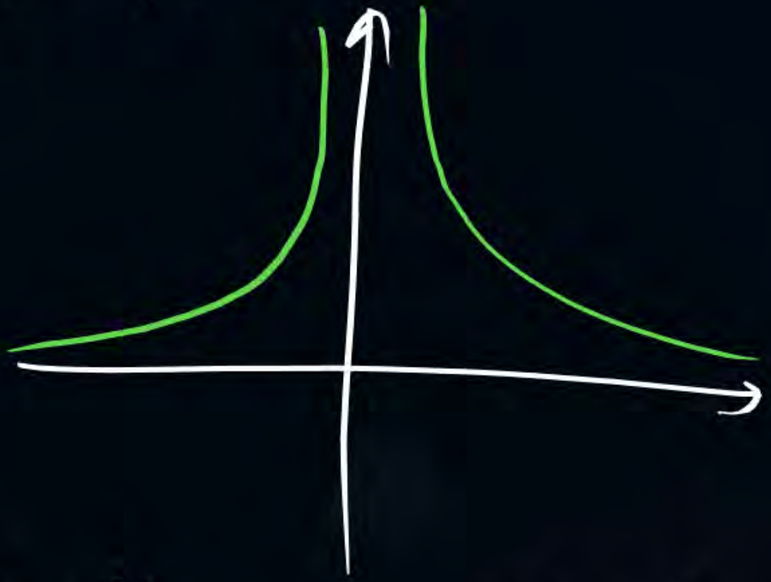
$y = x$, $y = x^2$, $y = x^3$



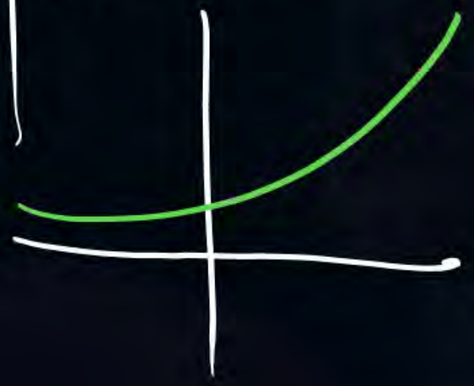
$y = \frac{1}{x}$, $y = \sqrt{x}$, $y = x^{1/3}$



$y = \frac{1}{x^2}$



$y = e^x$



$y = \log_e x$



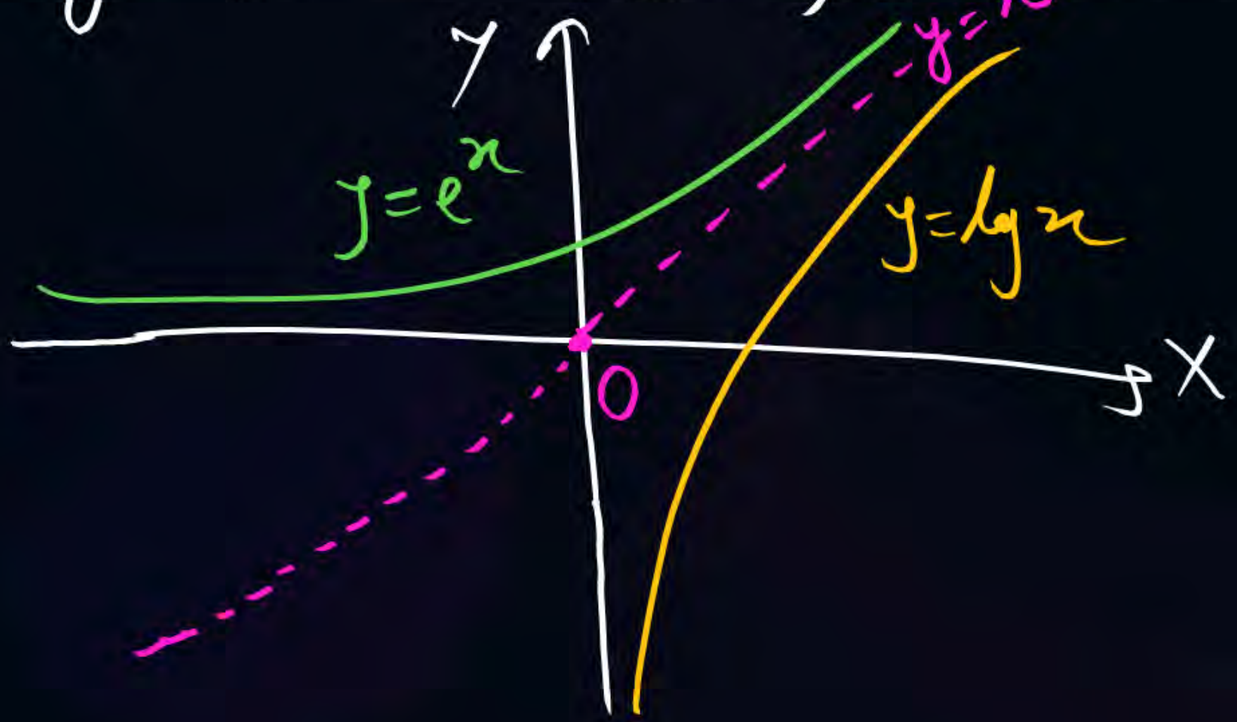
INVERSE func^n →

If $y=f(x)$ & $y=g(x)$ are Inverse func^n of each other

then they are symmetrical about the line $y=x$

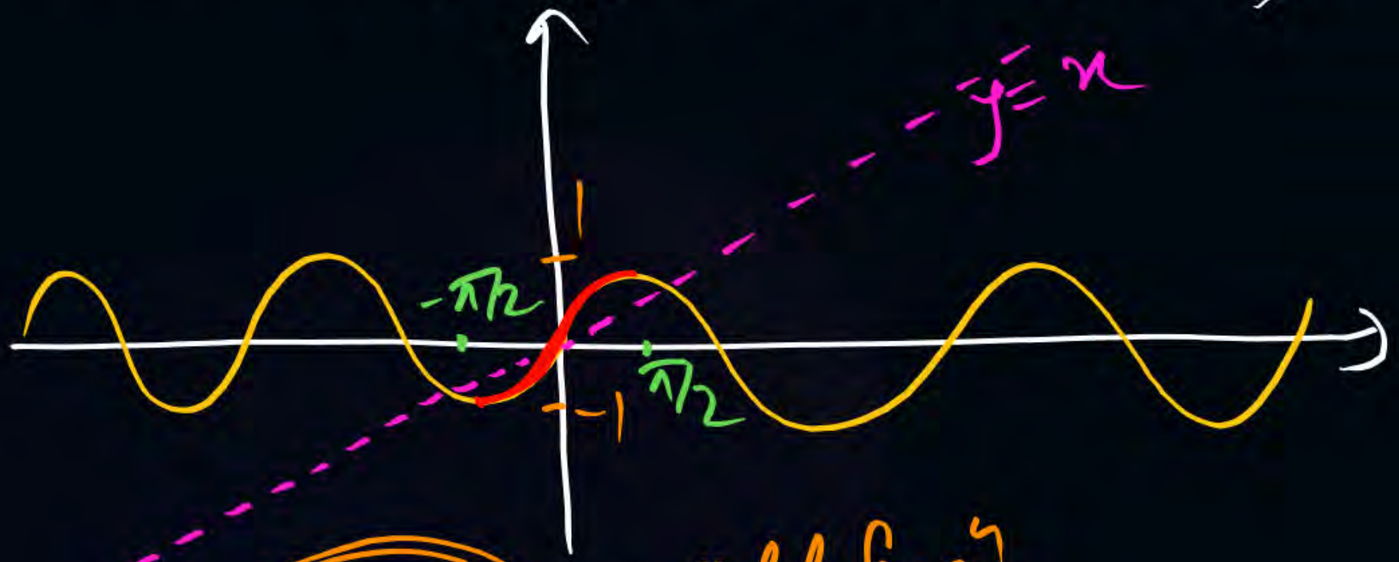
i.e. $y=x$ will behave like a mirror for $f(x)$ & $g(x)$.

eg $y=e^x$ & $y=\log e^x$ are Inverse func^n of each other



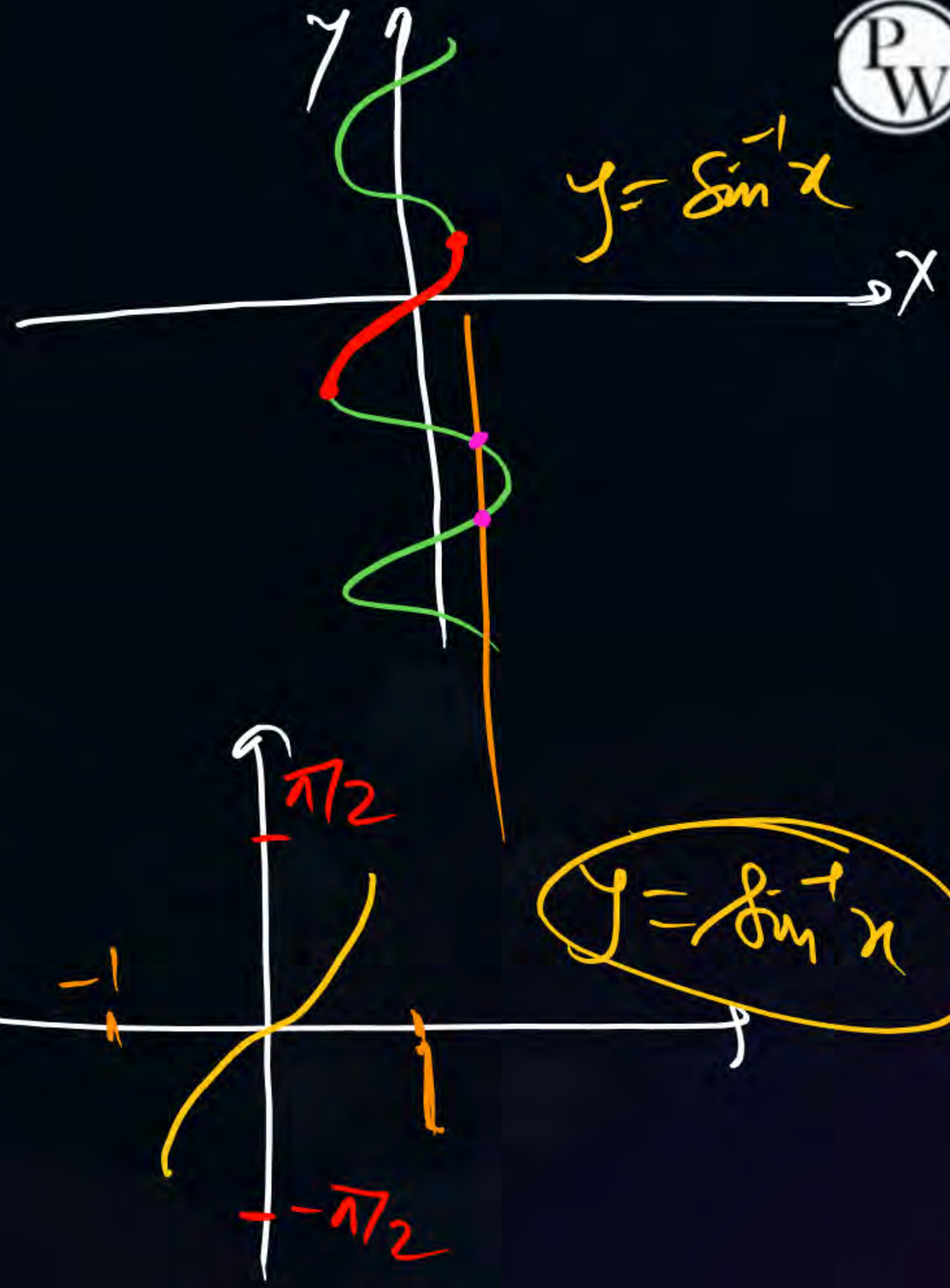
(*) $y = \sin^{-1} x$

We know that graph of $y = \sin x$ is as follows,



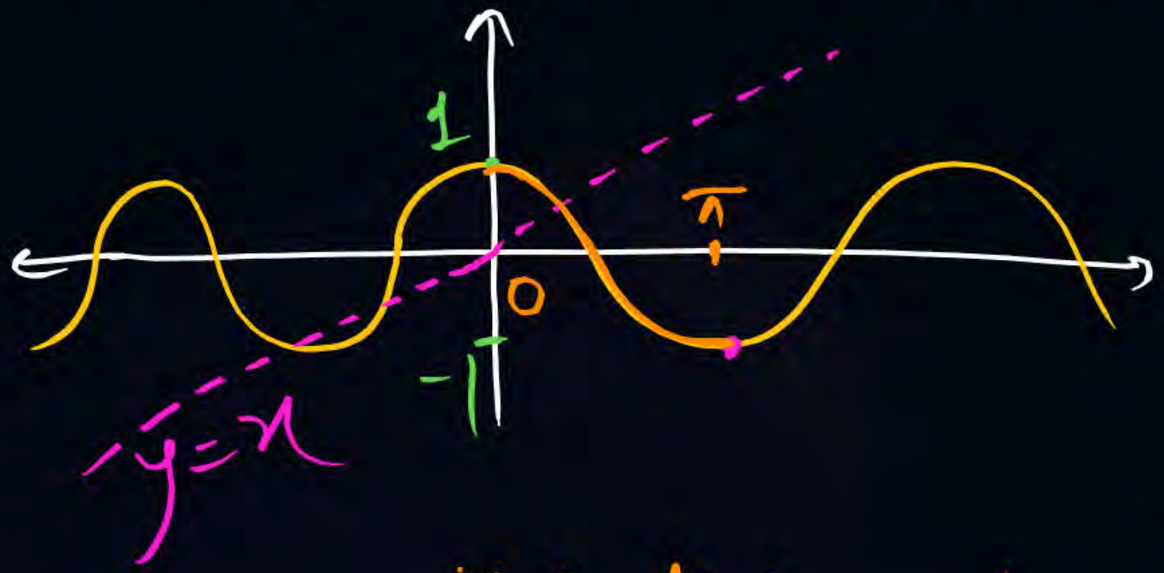
i.e. $y = \sin^{-1} x$

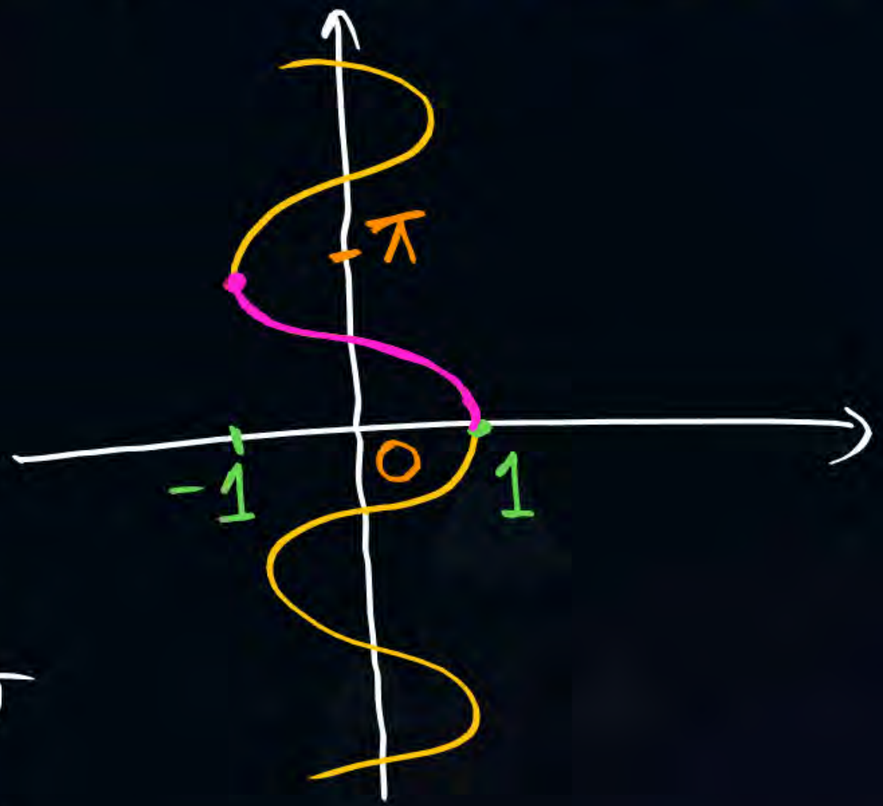
- odd func^n
- Domain $= [-1, 1]$
- Range $= \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$



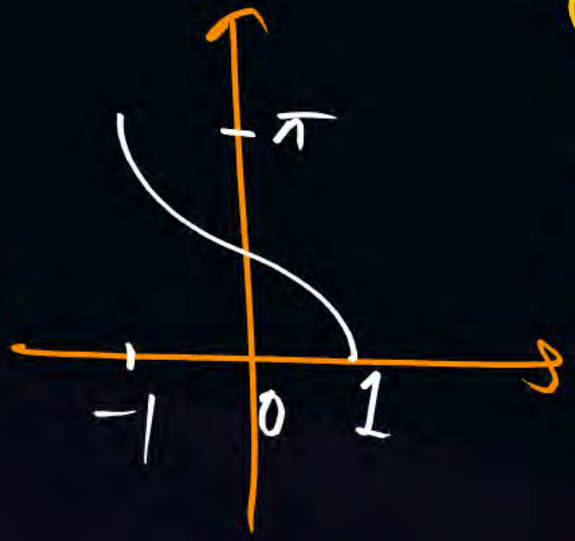
⊛ $y=\cos^{-1}x$; Domain $=[-1,1]$, Range $=[0,\pi]$, NEHO

w.k. that graph of $y=\cos x$ is as follows

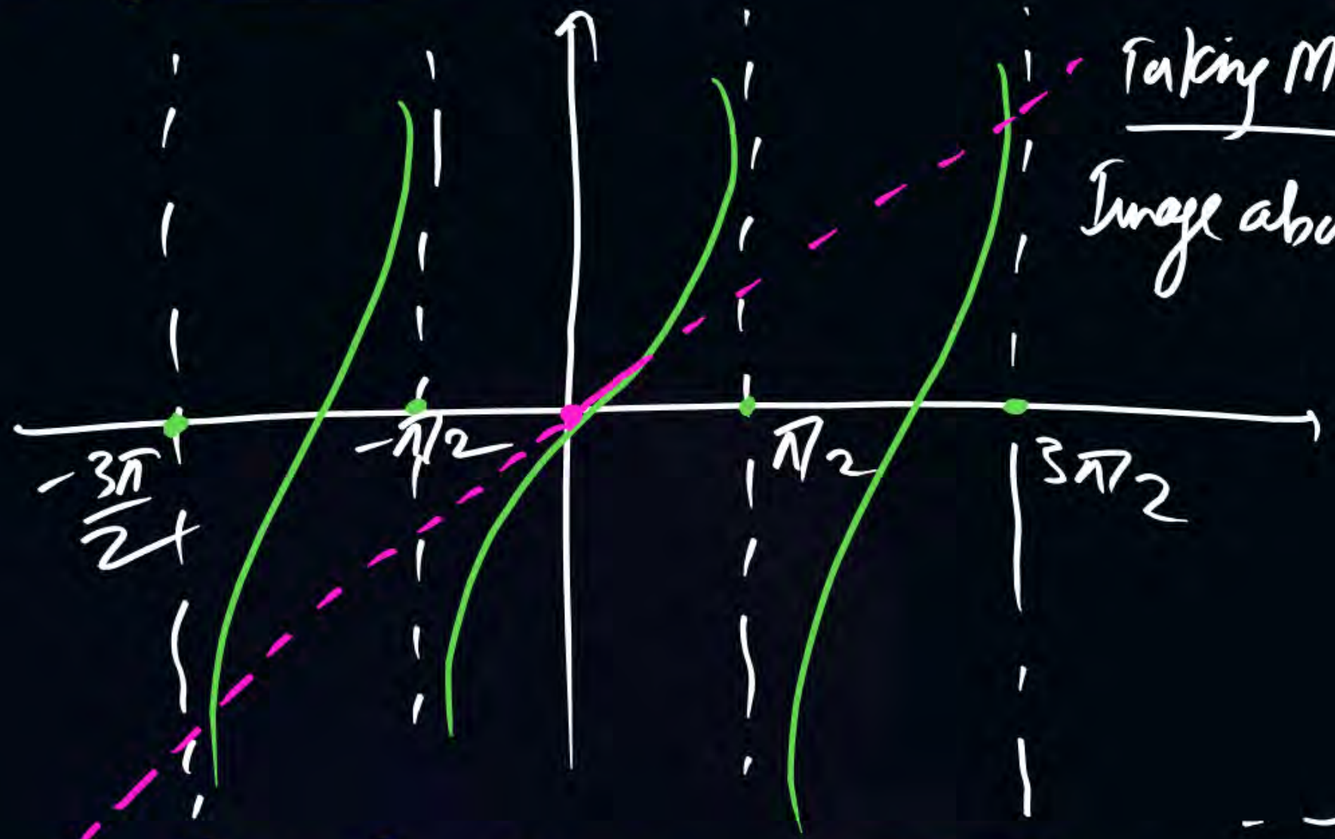




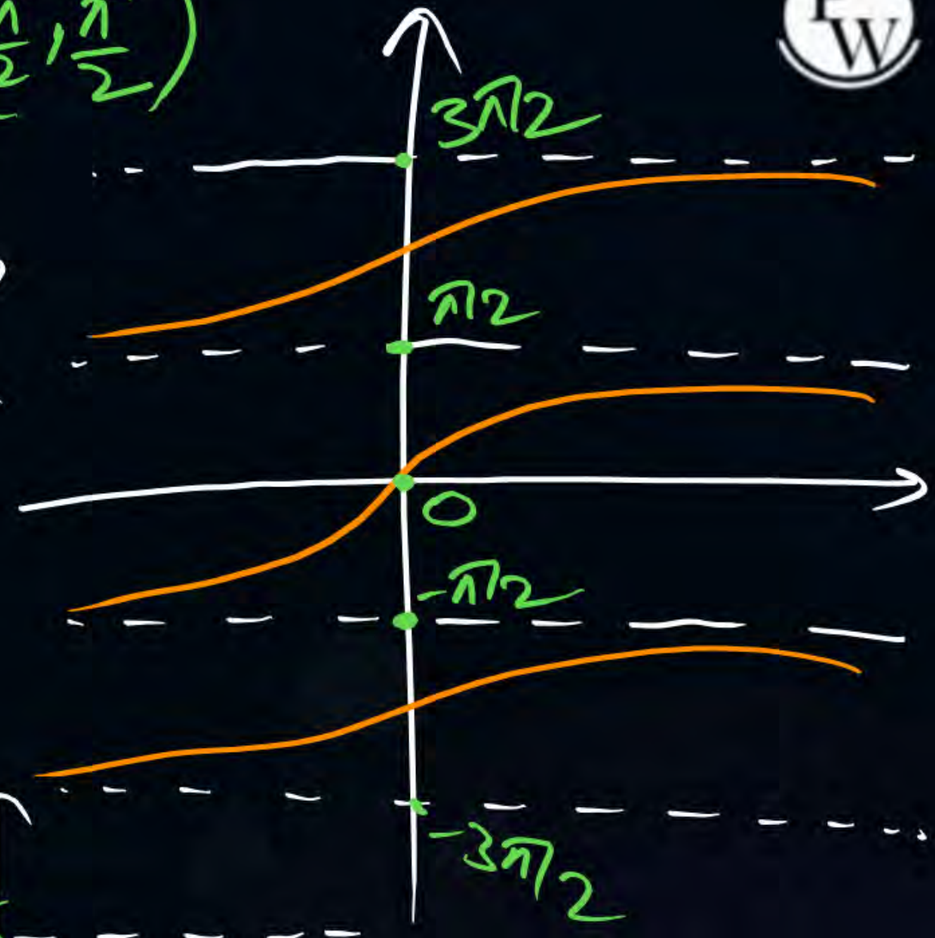
i.e graph of $y=\cos^{-1}x$ is



(x) $y = \tan^{-1}x$ Domain: $(-\infty, \infty)$; Range: $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$

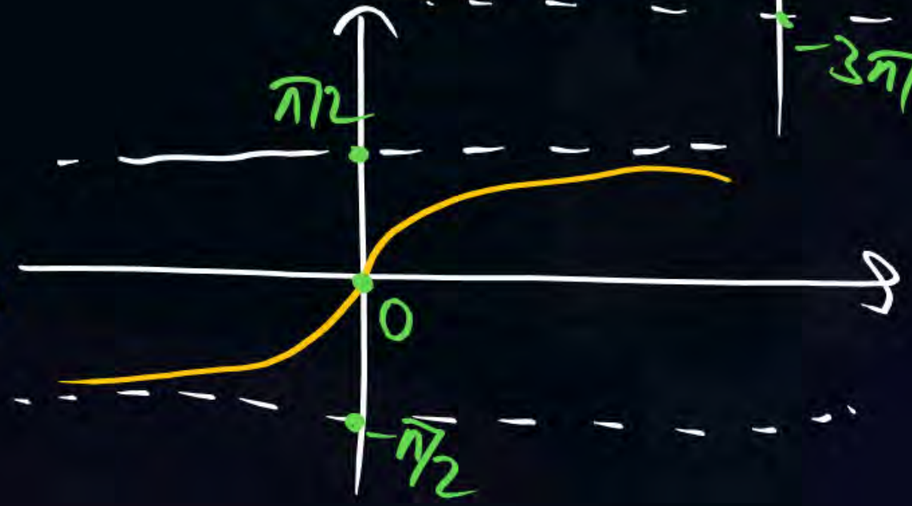


Taking Mirror Image about $y = x$



Hence Final Graph of $y = \tan^{-1}x$ is

& it is an ODD fnc^n



PIECEWISE func^n → If func^n is defined by Multiple sub function

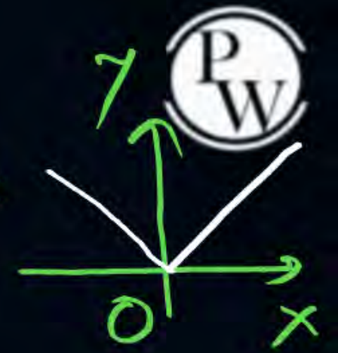
s.t, Domain of each subfunction is different

then function is called Piecewise func^n

for eg. Mod func^n, Signum func^n, G.I.F, L.I.F, fractional part func^n etc

Modulus func^n:

$$y = f(x) = |x| = \sqrt{x^2} = \begin{cases} -x, & x < 0 \\ +x, & x \geq 0 \end{cases} = \max(x, -x)$$



Domain $= (-\infty, \infty)$, Range $= [0, \infty)$, & it is an Even Function.

eg $|3| = +(3) = 3$

$|-3| = -(-3) = 3$

eg $|3| = \max(3, -3) = +3$

$|-3| = \max\{-3, -(-3)\}$

$= \max\{-3, 3\} = +3$

eg $|3| = \sqrt{(3)^2} = \sqrt{9} = +3$

$|-3| = \sqrt{(-3)^2} = \sqrt{9} = +3$

eg $x^2 - 9 = 0$

$x^2 = 9$

$x = \sqrt{9}$

$= \pm 3$ wrong step.

eg $x^2 - 9 = 0$

$x^2 = 9$

$x = \pm\sqrt{9}$

$= \pm(+3)$ Correct Version

$x = \pm 3$

Note Square Root of any Real Number is always +ve $\because \sqrt{x^2} = |x| = +ve$

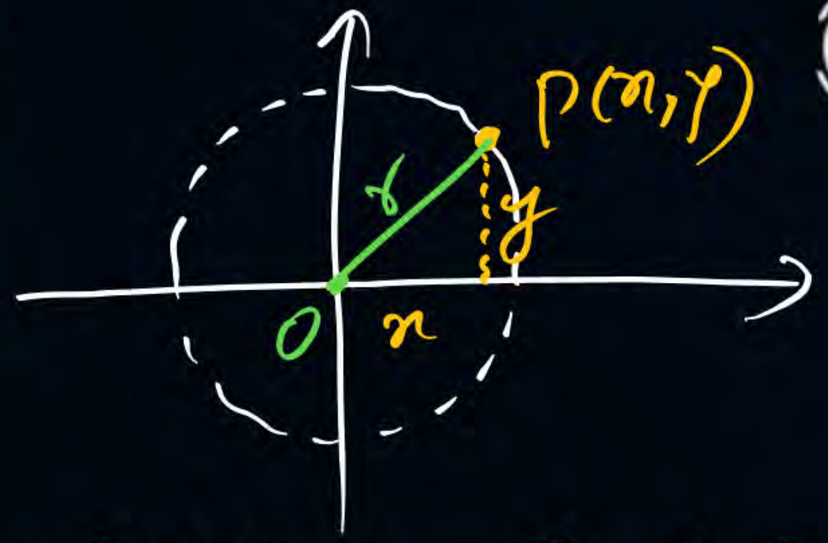
eg $[(-4)^2]^{\frac{1}{2}} = ?$ $[(-4)^2]^{\frac{1}{2}} = -4$ ✗

$\sqrt{(-4)^2} = \sqrt{16} = +4$ ✓

Special Note – we know that for Complex No.

$z = x + iy \Rightarrow r = OP = |z| = \sqrt{x^2 + y^2}$

$\because$ in Complex Analysis $|z|$ is the Representation of Radius & Radius can not be –ve so $|z|$ = always +ve



By Pythagoras Th, $x^2 + y^2 = r^2$

For Real z, Put $y = 0$ i.e

$z = x + i(0) \Rightarrow |z| = \sqrt{x^2 + 0^2}$

i.e $z = x$ or $|x| = \sqrt{x^2}$

$= +x$

Revision: $z = x + iy \Rightarrow |z| = \sqrt{x^2+y^2}$ = Radius

for Real Numbers, we have to put $y = 0$ i.e.

$\rightarrow z = x + i(0)$ & $|z| = \sqrt{x^2+0^2}$

or $z = x$ or $\boxed{|x| = \sqrt{x^2}}$ = +ve Number

Note $z = x + iy \Rightarrow |z| = \sqrt{x^2+y^2}$

for Purely Img No, Put $x = 0$

$z = 0 + iy$ & $|z| = \sqrt{0^2+y^2}$

i.e. $z = iy$ $\quad |iy| = \sqrt{y^2} \Rightarrow |i||y| = \sqrt{y^2} \Rightarrow |y| = \sqrt{y^2}$

Q: The Number of Solutions of $x^2-5|x|+6=0$ is/are ?= Four Sol?

Sol: Case I: if $x>0$ then

$$x^2-5(+x)+6=0$$
$$x^2-5x+6=0$$
$$x^2-2x-3x+6=0$$
$$x(x-2)-3(x-2)=0$$
$$(x-2)(x-3)=0$$
$$x=2,3$$

Case II: if $x<0$ then

$$x^2-5(-x)+6=0$$
$$x^2+5x+6=0$$
$$x^2+2x+3x+6=0$$
$$(x+2)(x+3)=0$$
$$x=-2,-3$$

Qe The Number of Solutions of $x^2+5|x|+6=0$ is/are ?= Zero Sol.

(M-I) (Using Common Sense) → (+ve No) + (+ve No) + (+ve No) = 0

i.e. AtQ, Sum of three +ve Nos = 0 ☹ ☹

i.e. Not Possible, So given eqn's has No Sol

(M-II) Case I: if $x>0$ then

$x^2+5(+x)+6=0$

$x^2+5x+6=0$

$x=-2,-3$

i.e. Not Possible

Case II if $x<0$ then

$x^2+5(-x)+6=0$

$x^2-5x+6=0$

$x=2,3$

i.e. Not possible

Some Useful Results :→

① $|x-a|<l \Rightarrow a-l<x<a+l$

eg $|x|<l \Rightarrow -l<x<l$

② $|x-a|>l \Rightarrow x<a-l$ or $x>a+l$

eg $|x|>l \Rightarrow x<-l$ or $x>l$

③ if $a<b$ s.t $(x-a)(x-b)<0$

$\Rightarrow a<x<b$

④ if $a<b$ s.t $(x-a)(x-b)>0$

$\Rightarrow x<a$ or $x>b$

Proof ① $|x-a|<l$

$\Rightarrow \pm(x-a)<l$

Case I:

$-(x-a)<l$

$(x-a)>-l$

$x>a-l$

Case II:

$+(x-a)<l$

$x<a+l$

or $a-l<x$; $x<a+l$

ie $a-l<x<a+l$

Hence Proved

Signum Function → it is defined as;

$$y = f(x) = Sgn(x) = \begin{cases} \frac{|x|}{x}, & x \neq 0 \\ 0, & x = 0 \end{cases} = \begin{cases} -1, & x < 0 \\ +1, & x > 0 \\ 0, & x = 0 \end{cases}$$



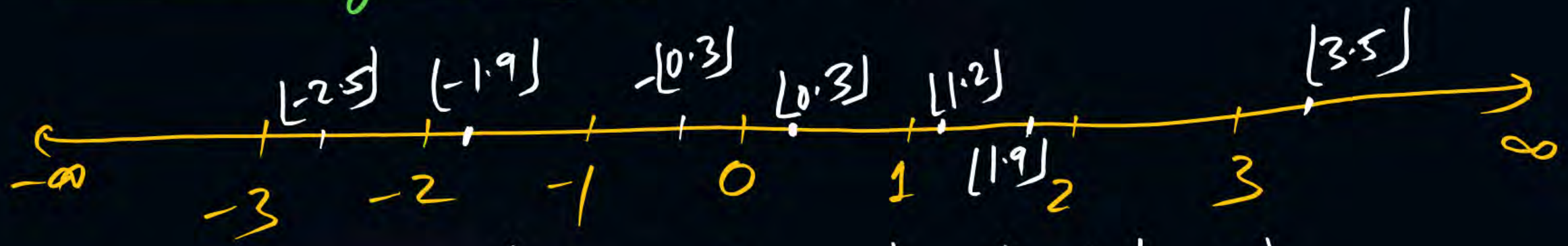
Domain = $(-\infty, \infty)$

Range = $\{-1, 0, 1\}$ is Range of Sgn Func^n Consisting only 3 Points +1, -1 & 0

& it is an ODD Func^n

eg $Sgn(0.4) = 1$, $Sgn(4.9) = 1$, $Sgn(-1.2) = -1$, $Sgn(0) = 0$ _ _ _ _ _

Greatest Integer func^n (Floor func^n) $\longrightarrow$



$\lfloor 0.3 \rfloor = 0$

$\lfloor -0.3 \rfloor = -1$

$\lfloor 1.2 \rfloor = 1$

$\lfloor 1.9 \rfloor = 1$

$\lfloor -1.9 \rfloor = -2$

$\lfloor 3.5 \rfloor = 3$

$\lfloor -2.5 \rfloor = -3$

$\lfloor 0 \rfloor = 0$

$\lfloor 1 \rfloor = 1$

$\lfloor 2 \rfloor = 2$

$\lfloor -1 \rfloor = -1$

$\lfloor -2 \rfloor = -2$

& so on----

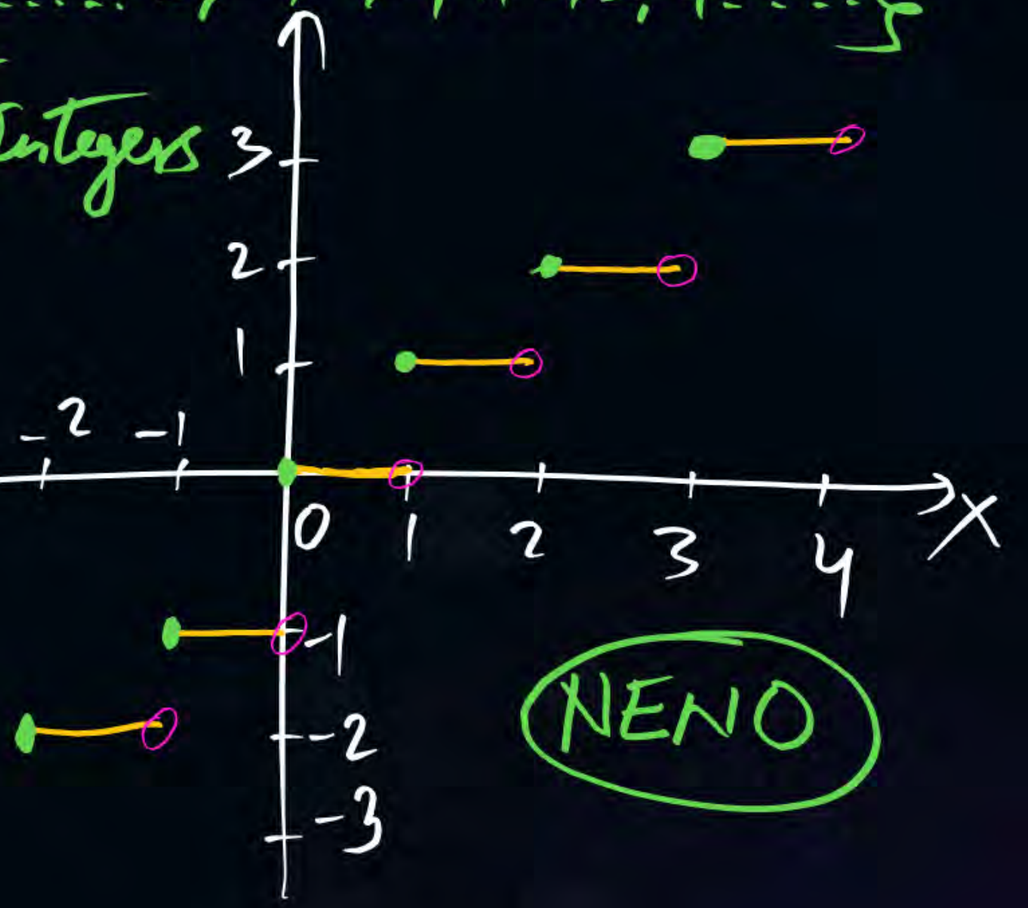
G.I.F: $y = f(x) = [x] = ?$

Domain $= (-\infty, \infty)$

Range $= \{\ldots -3, -2, -1, 0, 1, 2, 3, 4 \ldots\}$

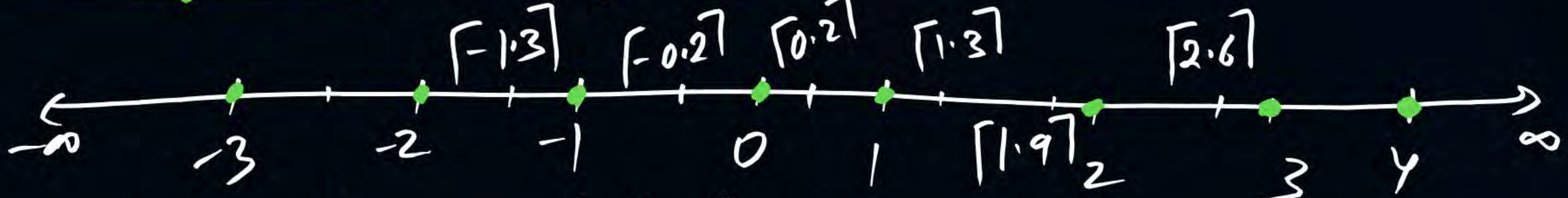
$=$ Set of Integers

$$y = [x] = \begin{cases} -2 & ; \ -2 \le x < -1 \\ -1 & ; \ -1 \le x < 0 \\ 0 & ; \ 0 \le x < 1 \\ 1 & ; \ 1 \le x < 2 \\ 2 & ; \ 2 \le x < 3 \\ 3 & ; \ 3 \le x < 4 \end{cases}$$



NENO

Least Integer Func^n: (Ceiling Func^n)



$\lceil 0.2 \rceil = 1$

$\lceil -0.2 \rceil = 0$

$\lceil 1.9 \rceil = 2$

$\lceil 1.3 \rceil = 2$

$\lceil -1.3 \rceil = -1$

$\lceil 2.6 \rceil = 3$

$\lceil 0 \rceil = 0$

$\lceil 1 \rceil = 1$, $\lceil -1 \rceil = -1$

$\lceil 2 \rceil = 2$, $\lceil -2 \rceil = -2$

Ref^{n} of Ceiling Function $y=\lceil x \rceil = ?$

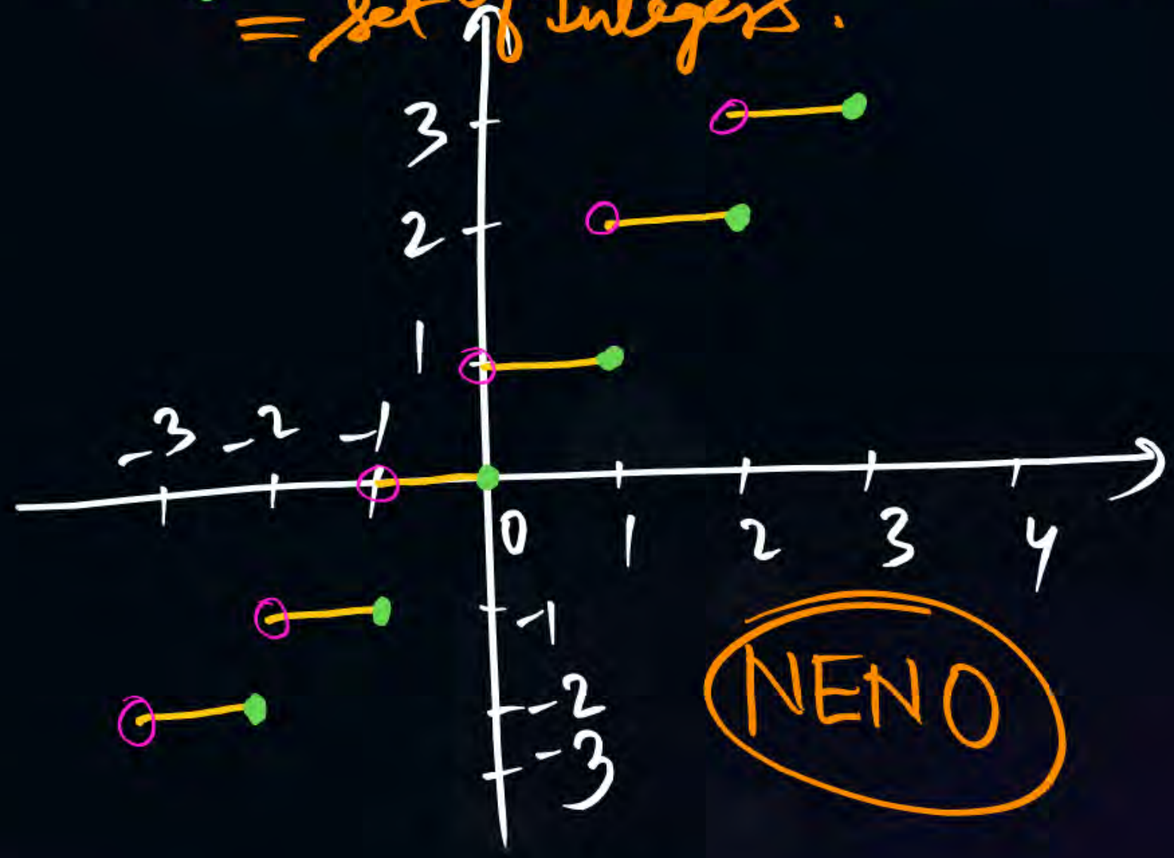
(C.I.F)

Domain $= (-\infty, \infty)$

Range $= \{\ldots -2, -1, 0, 1, 2, 3 \ldots\}$

$=$ set of Integers.

$$y=\lceil x \rceil = \begin{cases} \vdots & \vdots \\ -1 & , -2 < x \le -1 \\ 0 & , -1 < x \le 0 \\ \mathbf{1} & , 0 < x \le 1 \\ 2 & , 1 < x \le 2 \\ 3 & , 2 < x \le 3 \\ 4 & , 3 < x \le 4 \\ \vdots & \vdots \end{cases}$$



Fractional part func^n –

$$y = f(x) = \boxed{\{x\} = x - \lfloor x \rfloor}$$

(HW) Graph, Domain, Range, eg

? ? ? ?

THANK - YOU

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